EXPERIMENT 1.5

Mapped Course Outcomes-CO2

CO2: Analyze the different routing techniques.

Aim:- Implement Dynamic Routing using RIP (Routing Information Protocol)

Prerequisites:-

1. An access to CISCO PACKET TRACER .

RIP is a standardized Distance Vector protocol, designed for use on smaller networks. RIP was one of the first true Distance Vector routing protocols and is supported on a wide variety of systems. RIP has two versions, Version 1 (RIPv1) and Version 2 (RIPv2). RIPv1 (RFC 1058) is classful and thus does not include the subnet mask with its routing table updates. Because of this, RIPv1 does not support Variable Length Subnet Masks (VLSMs). When using RIPv1, networks must be contiguous, and subnets of a major network must be configured with identical subnet masks. Otherwise, route table inconsistencies (or worse) will occur. RIPv1 sends updates as broadcasts to address 255.255.255.255. RIPv2 (RFC 2543) is classless and thus does include the subnet mask with its routing table updates. RIPv2 fully supports VLSMs, allowing discontiguous networks and varying subnet masks to exist. [1]

RIP Characteristics.

· RIP sends out periodic routing updates (every 30 seconds)

· RIP sends out the full routing table every periodic update

· RIP uses a form of distance as its metric (in this case, hop count)

· RIP uses the Bellman-FordDistance Vector algorithm to determine the best “path” to a particular destination.

· RIP supports IP and IPXrouting.

· RIP utilizes UDP port 520.

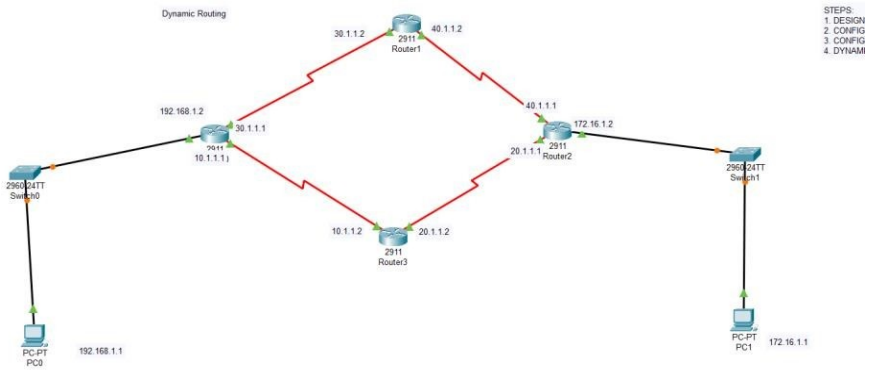
· RIP routes have an administrative distance of 120.

· RIP has a maximum hop count of 15 hops.

· Any network that is 16hops away or more is considered unreachable to RIP, thus the maximum diameter of the network is 15 hops. A metric of 16 hops in RIP is considered a poisoned route or infinity metric.

· If multiple paths exist to a particular destination, RIP will load balance between those paths (by default, up to 4) only if the metric (hop count) is equal. RIP uses an around-robin system of load-balancing between equal metric routes.

Topology.



Configuration.

Router0 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastEthernet 0/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 2/0

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 3/0

Router(config-if)#ip address 20.0.0.1 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#router rip

Router(config-router)#network 10.0.0.0

Router(config-router)#network 20.0.0.0

Router(config-router)#network 192.168.1.0

Router(config)#exit

Router1 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)# interface serial 2/0

Router(config-if)#ip address 10.0.0.2 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 3/0

Router(config-if)#ip address 30.0.0.1 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#router rip

Router(config-router)#network 20.0.0.0

Router(config-router)#network 30.0.0.0

Router(config)#exit

Router2 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)# interface serial 2/0

Router(config-if)#ip address 40.0.0.1 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 3/0

Router(config-if)#ip address 20.0.0.2 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#router rip

Router(config-router)#network 20.0.0.0

Router(config-router)#network 40.0.0.0

Router(config)#exit

Router3 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastEthernet 0/0

Router(config-if)#ip address 172.16.10.1 255.255.0.0 Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 2/0

Router(config-if)#ip address 40.0.0.2 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)# interface serial 3/0

Router(config-if)#ip address 30.0.0.2 255.0.0.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#router rip

Router(config-router)#network 40.0.0.0

Router(config-router)#network 30.0.0.0

Router(config-router)#network 172.16.0.0

Router(config)#exit

Further Reading:-

2. https://www.techopedia.com/definition/26161/static-routing

Research Articles

3. Dowdell, J. & Benamar, Nabil. (2015). Static Routing for DTN. Gardner, W.R. 1987. Water content: an overview. International Conference on Measurement of Soil and Plant Water Status. Centennial of Utah State Univ., pp. 7-9.

4. Saini, Himanshi and Pondwal, Veenu, An Analysis of Static and Dynamic Routing Techniques Based on Various Network Performance Parameters (May 2017). The IUP Journal of Telecommunications, Vol. IX, No. 2, May 2017, pp. 20-31, Available at SSRN: https://ssrn.com/abstract=3212546